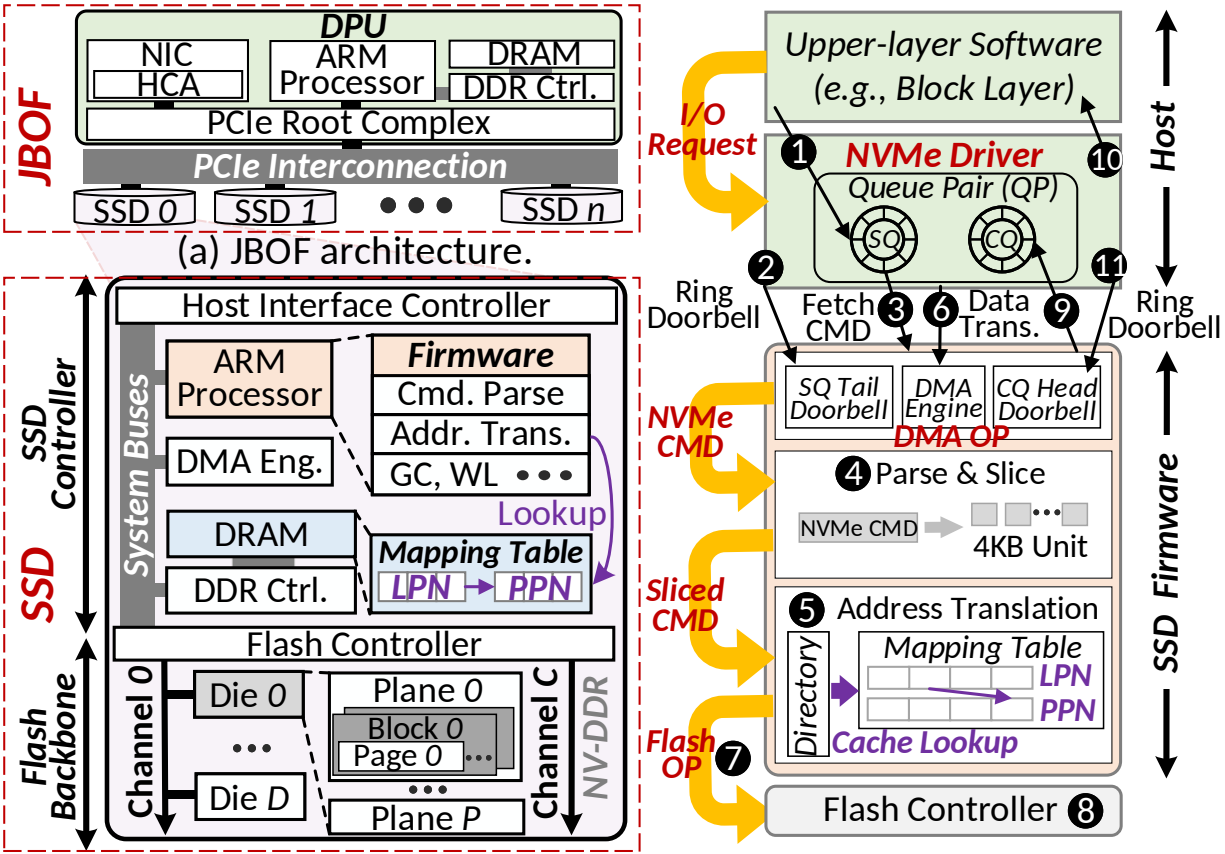

JBOF
DPU
NIC
HCA
ARM Processor
DRAM
DDR Ctrl.
PCIe Root Complex
PCIe Interconnection
SSD 0
SSD 1
SSD n
(a) JBOF architecture.
SSD
SSD Controller
Host Interface Controller
System Buses
ARM Processor
DMA Eng.
Firmware
Cmd. Parse
Addr. Trans.
GC, WL
Lookup
DRAM
DDR Ctrl.
Mapping Table
LPN
PPN
Flash Controller
Flash Backbone
Channel 0
Die 0
Die D
Plane 0
Block 0
Page 0
Plane P
Channel C
NV-DDR
(b) SSD architecture.
Upper-layer Software
(e.g., Block Layer)
I/O Request
NVMe Driver
Queue Pair (QP)
SQ
CQ
Host
Ring Doorbell
Fetch CMD
Data Trans.
Ring Doorbell
SQ Tail Doorbell
DMA Engine
CQ Head Doorbell
DMA OP
NVMe CMD
Parse & Slice
NVMe CMD
4KB Unit
Sliced CMD
Address Translation
Directory
Mapping Table
LPN
PPN
Cache Lookup
Flash OP
Flash Controller
SSD Firmware
(c) Details of I/O path.